Thermodynamic description of the Ni–Ti system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ni–Ti binary system as calculated with pycalphad from the SGTE dataset NiTi-04Tok.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Ni) + Ni3Ti, lies at 1575.6 K.

1. Introduction

The Ni–Ti system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, congruent compound. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of T. Tokunaga, K. Hashima, H. Ohtani, M. Hasebe, Mater. Trans. JIM, 45, 1507-14(2004). The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Ni reproduced by the dataset is 1728.2 K. For Ti, the liquidus meets the axis at 2498.8 K.

2. Phase diagram

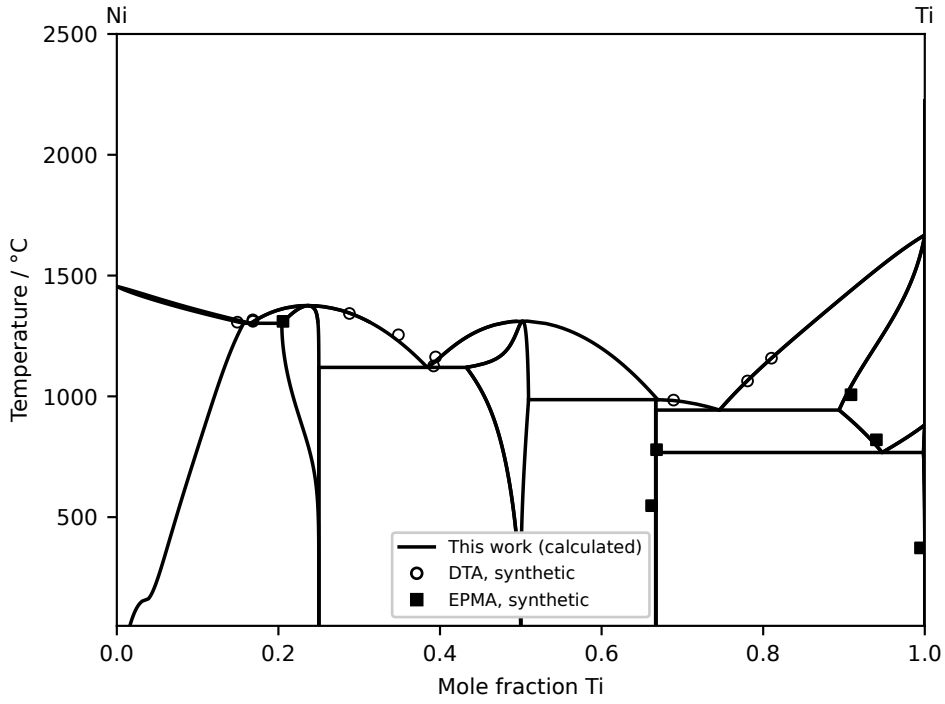


Fig. 1. Calculated Ni–Ti phase diagram with synthetic experimental-style points.

The dataset places a eutectic (L → (Ni) + Ni3Ti) at 1575.6 K; the composition of L is

16.6 at.% Ti. The eutectic reaction L → Ni3Ti + NiTi is calculated at 1393.0 K, with L at 38.4 at.%. We obtain 1259.5 K for the peritectic reaction (NiTi + L → NiTi2), where L has 66.9 at.% Ti. We obtain 1216.3 K for the eutectic reaction (L → NiTi2 + (βTi)), where L has 74.6 at.% Ti. We obtain 1040.8 K for the eutectoid reaction ((βTi) → NiTi2 + (αTi)), where (βTi) has 94.7 at.%. Ni3Ti melts congruently at 1649.4 K and 23.7 at.% Ti. NiTi2 melts congruently at 1259.5 K and 66.7 at.% Ti.

Table 1. Invariant reactions in the Ni–Ti system (calculated).

Reaction	Type	T / °C	Compositions (x(Ti))		
			phase 1	phase 2	phase 3
L → (Ni) + Ni3Ti	eutectic	1302.5	0.166	0.157	0.204
L → Ni3Ti + NiTi	eutectic	1119.9	0.384	0.250	0.433
NiTi + L → NiTi2	peritectic	986.4	0.510	0.669	0.667
L → NiTi2 + (βTi)	eutectic	943.2	0.746	0.667	0.894
(βTi) → NiTi2 + (αTi)	eutectoid	767.6	0.947	0.667	0.998

3. Thermodynamic properties

At 2600 K the integral mixing enthalpy of the liquid has its minimum, −40.84 kJ/mol, at 38.0 at.% Ti. The calculated enthalpy of formation of NiTi2 at 298.15 K is −26.90 kJ/mol per mole of atoms.

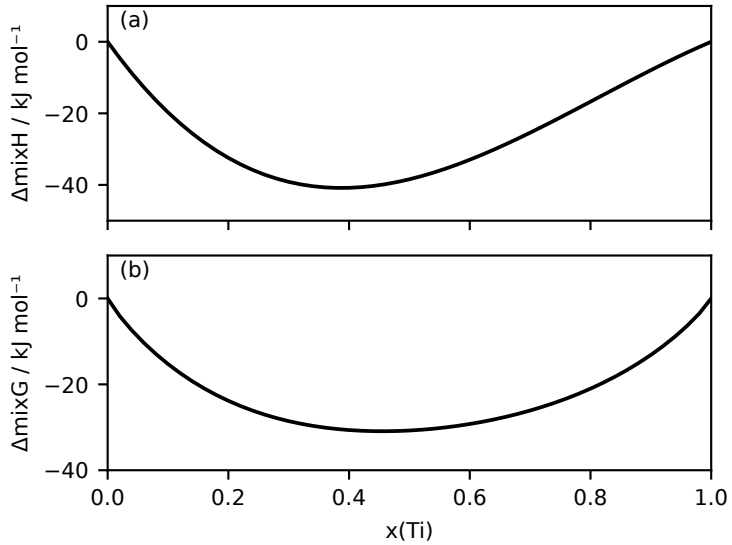


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 2600 K.

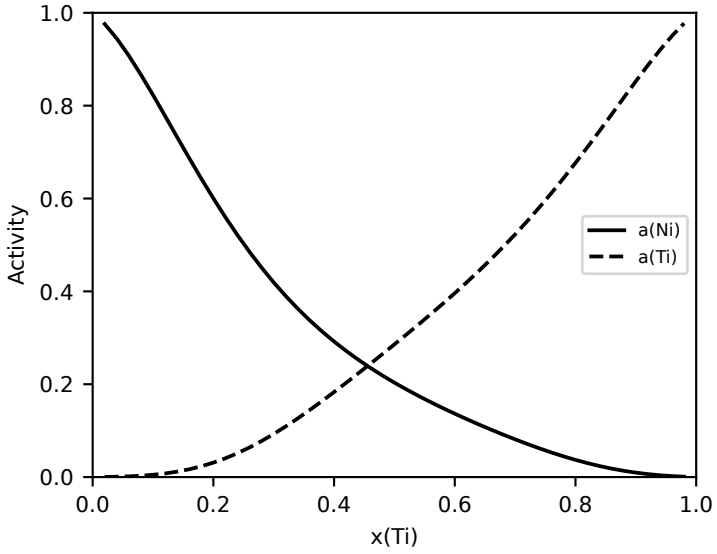


Fig. 3. Calculated activities in the liquid at 2600 K (reference: pure liquids).

Table 2. Calculated formation properties of the intermediate phases (Ni–Ti).

		Calculated at 298.15 K, per mole of atoms	
Phase	**at.% Ti**	**ΔfH / kJ mol⁻¹**	**S / J mol⁻¹ K⁻¹**
Ni3Ti	25.0	-40.04 ± 0.5	25.09
NiTi2	66.7	-26.90 ± 2.0	28.63
NiTi	49.9	-33.34 ± 0.5	31.48

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.